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Highlights

- Firms facing higher biodiversity risk tend to increase corporate investment.
- The positive relationship between biodiversity risk and corporate investment is more pronounced in highly competitive industries.
- Strategic preemptive behavior driven by competitive pressures primarily explains this positive association.
- The positive association between biodiversity risk and investment is amplified by higher climate policy uncertainty but weakened by managerial myopia.

Threat or opportunity: Biodiversity risk and corporate investment

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Abstract: We investigate how biodiversity risk influences corporate investment. Using text-based measures of firm-level exposure, we find that higher biodiversity risk is associated with significantly greater investment. This effect is more pronounced in highly competitive industries, consistent with the strategic growth option view: when first-mover advantages matter, firms may respond to biodiversity risk by investing early. The effect is amplified for firms facing volatile external regulatory environment, but attenuated in firms with more managerial myopia. The results are robust across alternative specifications and endogeneity checks, highlighting biodiversity risk as an economically meaningful and independent driver of corporate investment decisions.

Keywords: Biodiversity risk, Corporate investment, Real option theory, Strategic growth option

1. Introduction

Biodiversity loss imposes sizable economic costs. For instance, Frank and Sudarshan (2024) estimate that India's vulture population collapse caused USD 69.4 billion in annual losses due to increased human mortality. Such evidence underscores that biodiversity risk extends beyond ecological harm to influence economic outcomes and corporate strategy. Recent studies link biodiversity risk to corporate cash holdings, dividends, financing, and bankruptcy (Ahmad and Karpuz, 2024; Zhou et al., 2025; Liu et al., 2025; Li et al., 2025; Adamolekun, 2024), reinforcing the view that firms respond to such risk with caution. Yet an alternative possibility remains

underexplored: some firms may treat biodiversity risk as a strategic opportunity, anticipating future regulation or demand shifts. Proactive investment in eco-friendly technologies may turn a looming liability into a competitive edge (Kulatilaka and Perotti, 1998). This tension raises a key question at the intersection of biodiversity risk and corporate finance: does biodiversity risk ultimately deter or encourage corporate investment?

Biodiversity risk, defined as the potential financial and operational losses firms face when ecosystem functions deteriorate (OECD, 2023), is pervasive yet poorly understood and largely unpriced, making it a latent source of uncertainty for corporate decision-makers. Because the timing and impact of biodiversity threats are difficult to predict, one manager may perceive it as a threat, whereas another might see it as a window of opportunity. Real options theory offers two opposing predictions. The first is the classic “delay option” argument (Bernanke, 1983; Dixit and Pindyck, 1994). It holds that greater uncertainty increases the value of waiting for new information, thereby suppressing immediate investment. In contrast, the “strategic growth option” perspective (Kulatilaka and Perotti, 1998) argues that when first-mover advantages are significant, uncertainty can instead spur early investment commitment.

In our empirical analysis, using panel data on 5,000 Chinese listed firms from 2007 to 2023, we find that greater biodiversity risk exposure is associated with higher corporate investment. The effect is more pronounced in highly competitive industries, consistent with the strategic growth option mechanism. In other words, managers appear to interpret biodiversity risk as an opportunity to secure future competitive rents, contrasting with the prevailing “risk-only” narrative surrounding biodiversity risk (Ahmad and Karpuz, 2024; Zhou et al., 2025; Liu et al., 2025; Li et al., 2025; Adamolekun, 2024).

This paper makes three contributions. First, we provide novel evidence that biodiversity risk promotes corporate investment, suggesting that firms interpret biodiversity exposure not merely as a hazard but as a latent opportunity. This finding offers empirical support for the strategic growth option view of real options theory, whereby uncertainty—when coupled with potential first-mover advantages—can stimulate investment (Kulatilaka and Perotti, 1998; Folta and O’Brien, 2004). Second, we uncover significant heterogeneity in firms’ responses to biodiversity risk: the investment-promoting effect is more pronounced in firms facing volatile external regulatory environment and less so in firms with more managerial myopia. This highlights the importance of both external policy environment and internal governance in shaping how firms interpret and act upon biodiversity-related risks. Third, we broaden the literature on corporate investment by identifying biodiversity risk as a novel, non-traditional determinant of corporate investment (Gulen and Ion, 2016).

2. Literature review and hypothesis development

When assessing the environmental economic impact, attention has centered on climate risk (Ginglinger and Moreau, 2023; He et al., 2025a; Li et al., 2024; Pankratz et al., 2023; Zhang and Kong, 2025), while an equally critical parallel challenge is emerging: biodiversity risk. Unlike climate risks, which are linked to the negative consequences of climate change and manifest as long-term, systemic changes, biodiversity risks stem from the disruption of living systems. These processes directly impair essential ecosystem services like pollination, water regulation, soil stability, and disease control, which are critical for the operational continuity of many industries, such as agriculture, forestry, fisheries, and pharmaceuticals. The breakdown of these services can

cause abrupt and localized shocks that severely disrupt supply chains, reduce the availability and quality of raw materials, and compromise production capacity (Duong et al., 2025).

Recognizing that climate and biodiversity risks are connected but distinct, a growing body of research has emerged to measure its impact. In particular, Giglio et al. (2023) and He et al. (2024) pioneered firm-level exposure indices for the United States and China, respectively, laying the empirical groundwork for subsequent analyses. Building on these data innovations, a growing literature links rising biodiversity risk exposure to firms' cash policies, payout behavior, financing costs, digital transformation, and financial distress (Ahmad and Karpuz, 2024; Zhou et al., 2025; Liu et al., 2025; Wang, 2025; Li et al., 2025). However, no prior study has examined how biodiversity exposure influences corporate investment. We fill this gap, advancing our understanding of the effects of biodiversity risk.

We propose hypotheses from two opposing perspectives. When investment is partially irreversible, greater uncertainty increases the value of waiting (Bernanke, 1983; Dixit and Pindyck, 1994). This classic real-options argument implies that heightened biodiversity risk could deter immediate investment as firms exercise the option to delay. By deferring projects until conditions are clearer, managers preserve the flexibility to invest later and avoid the potential costs of committing too early. Thus:

H1a. Greater exposure to biodiversity risk reduces corporate investment.

By contrast, competitive dynamics can erode the value of waiting. Rivalry may undermine the benefits of delay because early commitment can secure pre-emptive advantages—such as market share, technological leadership, or regulatory goodwill—that competitors cannot easily replicate (Grenadier, 2002). Kulatilaka and Perotti (1998) formalize this mechanism as a strategic growth option, demonstrating that uncertainty can increase investment incentives when first-mover gains outweigh the option value of waiting. Biodiversity risk could create precisely such conditions by signaling future environmental regulation or shifting consumer preferences toward eco-friendly products, thereby rewarding firms that invest proactively. Thus:

H1b. Greater exposure to biodiversity risk increases corporate investment.

3. Data and methodology

3.1 Biodiversity risk and climate risk measures

Due to a lack of firm-level data, studies on biodiversity and corporate decision-making are currently limited. Following Giglio et al. (2023), He et al. (2024) apply a text analysis approach to Chinese A-share firms' annual reports from 2007 to 2022, counting the frequency of biodiversity-related terms (e.g., biodiversity, ecosystem, habitat, and species). They construct three measures: (1) the Biodiversity Concern Index (*Bio_Risk*)—the ratio of biodiversity-related keyword characters to the total characters in the report, capturing length-normalized intensity; (2) the Biodiversity Keyword Count (*Alt_Bio_Risk1*)—the raw frequency of dictionary terms; and (3) the Biodiversity Risk Index (*Alt_Bio_Risk2*)—a binary indicator equal to 1 if biodiversity terms appear more than twice, and 0 otherwise. We use *Bio_Risk* as our primary explanatory variable and the latter two as alternative measures for robustness.

Because biodiversity loss is closely linked to climate change, we include two province-level climate risk controls to isolate biodiversity-specific effects. We include climate physical risk (*Climate_Risk_P*; Guo et al., 2024) and climate policy uncertainty (*CPU_P*; Ma et al., 2023). By

accounting for both physical and transition climate risks, we ensure that the estimated effect of biodiversity risk is not confounded by climate risk exposure.

3.2 Data and regression model

Our sample comprises all non-financial A-share companies listed in China from 2007 to 2023. After excluding financial institutions, ST (Special Treatment) firms, and observations with missing data, the final sample contains 5,000 firms and 39,454 firm-year observations. Financial-statement and governance variables come from CSMAR. The relationship is estimated with the following fixed-effects panel specification:

$$Invest_{i,t} = \alpha_i + \beta_1 Bio_Risk_{t-1} + \beta_2 Climate_Risk_P_{i,t-1} + \beta_3 CPU_P_{i,t-1} + \beta_4 Controls_{i,t-1} + Firm_i + Year_{t-1} + \varepsilon_{i,t} \quad (1)$$

where *Invest* is the investment ratio of firm *i* in year *t*. The key independent variable, *Bio_Risk*, captures firm-level biodiversity exposure, while *Climate_Risk_P* and *CPU_P* proxy for climate physical risk and transitional risk, respectively. One-year-lagged controls include Tobin's Q, sales growth, cash flow, irreversibility, size, leverage, regional GDP, board male ratio, and board average age. All continuous variables are winsorised at the 1st and 99th percentiles. Definitions of all variables and descriptive statistics are presented in Tables 1 and 2.

[Insert Table 1 here]

[Insert Table 2 here]

4. Empirical results

4.1 Baseline regression results

Table 3 presents the baseline fixed-effects regression estimates. Biodiversity risk exposure is positively associated with corporate investment across all model specifications. In our full model, the coefficient on *Bio_Risk* is 7.0188 ($t = 3.52$), significant at the 1% level. This effect is economically meaningful: a one-standard-deviation increase in *Bio_Risk* is associated with roughly a 0.0028 units (7.0188×0.0004) increase in the investment ratio, equivalent to 4.63% of the sample mean.

[Insert Table 3 here]

Importantly, the positive biodiversity-investment relationship persists after controlling for climate risks. Climate physical risk (*Climate_Risk_P*) negatively affects investment, while climate policy uncertainty (*CPU_P*) shows no significant effect. This suggests biodiversity risk captures a distinct investment driver. Overall, the baseline evidence supports Hypothesis H1b: managers appear to treat biodiversity risk as a strategic growth opportunity rather than as a reason to defer investment.

4.2 Mechanism

Strategic growth option models suggest that intense product-market rivalry erodes the value of waiting, pushing firms toward preemptive investment (Kulatilaka and Perotti 1998; Grenadier 2002). To test this mechanism, we examine whether the investment-promoting effect is amplified in more

competitive industries. Table 4 shows regressions splitting the sample by competition intensity. We use three proxies for competition: the Herfindahl-Hirschman Index (HHI), the four-firm concentration ratio (CR4), and the eight-firm concentration ratio (CR8). Industries are grouped annually into “high” and “low” competition categories based on each proxy’s mean, and Equation (1) is re-estimated.

Across all competition metrics, biodiversity risk remains positively associated with investment, but effects are stronger under high competition. Under the HHI split, the Bio_Risk coefficient is 9.6440 ($t=3.15$) for high-competition subsample, versus 5.8237 ($t=2.16$) for low-competition subsample, with a statistically significant difference ($p=0.010$). The CR4 and CR8 splits yield similar patterns: coefficients in high-competition subsamples are 9.9231 ($t=3.36$) and 8.9215 ($t=3.05$), respectively, compared to 5.4716 ($t=1.95$) and 5.2090 ($t=1.85$) in low-competition settings, with significant differences at the 10% levels. The difference in coefficients between the high- and low-competition subsamples is statistically significant ($p=0.010$ and $p=0.000$). These results are consistent with the strategic growth option mechanism (Grenadier, 2002; Kulatilaka and Perotti, 1998): greater rivalry increases the payoff to moving first, so firms facing high biodiversity risk in competitive industries accelerate capital spending rather than wait. Such behavior stands in contrast to the delay-option prediction, which would imply a muted or negative investment response to heightened uncertainty.

[Insert Table 4 here]

[Insert Table 5 here]

To provide further evidence for this mechanism, we conducted additional tests in Table 5. Based on the preceding theory, we argue that firms' proactive pursuit of strategic growth should be reflected in their concrete strategic behaviors. Therefore, we used firms' green innovation patents and green utility models to measure firms' concrete strategic actions. The results reveal a significant positive relationship between corporate biodiversity risk exposure and green patents (189.2324 and 151.2053 with t -values of 5.75 and 5.02). This finding explicitly demonstrates that firms are responding proactively rather than merely complying with regulations, which strengthens the interpretation of biodiversity risk as an opportunity. In sum, the mechanism analysis and our further analysis on green innovation reinforce the conclusion that managers predominantly view biodiversity risk as a window for preemptive growth, not merely as a threat to be avoided.

4.3 Robustness

We conduct additional tests to confirm that our findings are not artifacts of specific measurement or sampling choices. First, we verify the positive biodiversity-investment link under alternative variable definitions. Table 6, columns (1)-(3), replace Bio_Risk with a word-count measure (*Alt_Bio_Risk1*) and a binary exposure indicator (*Alt_Bio_Risk2*); column (3) also uses an alternative dependent variable (*Alt_Invest*; see Table 1 for definitions). Across all cases, biodiversity-risk coefficients remain positive and significant, and the climate-risk controls behave consistently, confirming the robustness of our result.

[Insert Table 6 here]

Second, to mitigate sample selection bias, we perform entropy balancing in column (4). We classify a firm-year as “exposed” if its annual report contains more than two biodiversity-related

keywords (*Alt_Bio_Risk2*), then reweight the observations so that exposed and unexposed firms have similar observable characteristics. The coefficient on *Bio_Risk* in the balanced sample remains positive and significant. Together, these robustness checks strengthen our conclusion that managers treat biodiversity risk as a growth opportunity rather than as a signal to postpone investment.

Third, to address endogeneity, we implement a two-stage least squares (2SLS) regression using two instrumental variables (IV): (i) the average *Bio_Risk* of same-industry firms in other provinces, and (ii) the province-level forest coverage rate. Following the approach of He et al. (2025b) in their selection of natural environment indicators for IV, we use provincial forest coverage as a valid instrument. Provincial forest coverage, as a vital component of regional ecosystems, is not only intrinsically linked to biodiversity but also acts as a key performance indicator for local government ecological efforts, indirectly influencing how firms perceive biodiversity risks. Importantly, this variable is largely determined by geographical and natural factors, making it exogenous to firm-specific decision-making and thus satisfying the exogeneity and relevance conditions. To address the concern that forest coverage may correlate with regional economic activity, our model controls for this relationship by including province-level GDP, as well as firm and year fixed effects.

The other instrument, the biodiversity risk exposure of same-industry firms in other provinces, is correlated with a firm's own exposure through shared industry shocks and information flows, yet it does not directly affect the firm's investment choices. In the first stage (Table 6, column 5), both instrument variables are all significant. In the second stage (Table 6, column 6), the *Bio_Risk* coefficient remains positive and significant (31.0634, $t=1.98$). The Kleibergen-Paap Wald F-statistic is 23.13 (above the 10% threshold), and the Hansen J-test p-value is 0.2154 (>0.1), confirming instrument strength and validity. We further employ two additional instrument variables: (i) the annual average biodiversity risk at the industry level; and (ii) the average biodiversity risk of industry peers located within the same three-digit ZIP code area. Our findings remain robust after conducting these tests.

To provide further causal evidence for the effect of biodiversity risk, we employ a difference-in-differences (DID) framework. Following Zhou et al. (2025), we leverage the 2011 China Biodiversity Conservation Strategy and Action Plan, a policy that designated 35 biodiversity priority areas to guide conservation efforts. This policy created an exogenous shock that increased the regulatory exposure and social expectations for firms in the designated areas. Taking 2011 as the policy year, we define firms in priority provinces as the treatment group and others as the control. As shown in Table 6 (column 7), the DID regression results reveal a significant positive effect of the policy on corporate investment (0.0132, $t=2.72$), providing strong causal evidence that is consistent with our main findings.

5. Further analysis

The impact of biodiversity risk on investment may be shaped by external regulatory environment and firm-specific governance structures. Accordingly, we perform heterogeneity analyses along the dimensions of climate policy uncertainty and managerial myopia.

First, we use the province-level climate policy uncertainty (*CPU_P*) index to proxy for external regulatory environment. Firms in high-*CPU_P* regions face greater policy uncertainty, which may prompt them to interpret biodiversity risk as a forward-looking signal and invest preemptively to secure strategic advantage (Grenadier, 2002). In Table 7 column (1)-(2), we split the sample at the mean value of *CPU_P* and find that the effect of biodiversity risk on investment is significantly

stronger in high-CPU_P regions: the coefficients are 5.4228 ($t=1.75$) and 9.6412 ($t=3.42$) for low- and high-CPU_P regions, respectively, with a significant difference ($p=0.010$). This suggests that firms in more uncertain regulatory environments are more likely to treat biodiversity risk as a strategic opportunity.

[Insert Table 7 here]

Second, we further explore internal governance by assessing whether biodiversity risk affects investment differently depending on managerial myopia. Myopic managers tend to underinvest in initiatives with long-term payoffs, including responses to environmental risks. In contrast, forward-looking managers are more likely to view biodiversity risk as a strategic signal and invest preemptively (Zhao et al., 2025). Using myopia index developed by Hu et al. (2021), we find that the *Bio_Risk* coefficient is 7.3877 ($t=3.01$) for low-myopia firms and 4.5856 ($t=1.51$) for high-myopia ones, with a statistically significant difference ($p=0.050$), shown in Table 7 column (3)-(4). The results indicate that forward-looking managers are more inclined to treat biodiversity risk as a growth opportunity, aligning with the strategic growth option theory.

6. Conclusions

We present new and robust evidence that a firm's biodiversity risk exposure significantly raises corporate investment. The effect is most pronounced in industries where product-market competition is intense, a setting that amplifies first-mover advantages and erodes the option to wait, proving the strategic growth option theory. Extensive robustness check-alternative variables, IV estimation, DID estimation, and entropy balancing-confirm that the result is not driven by measurement or endogeneity concerns. Furthermore, heterogeneity analyses reveal that the investment-promoting effect of biodiversity risk is stronger for firms facing greater climate policy uncertainty and those with less managerial myopia. These findings suggest that both volatile external regulatory environment and long-term-oriented internal governance reinforce the strategic interpretation of biodiversity risk.

Our findings also offer crucial insights into both managerial and policy implications. For managers, this research highlights the importance of a long-term strategic outlook. While myopic managers often avoid investment to secure short-term gains, forward-looking leaders can transform these risks into opportunities and gain a crucial first-mover advantage in a shifting market. To better align corporate strategy with environmental stewardship, it is essential for corporate boards to adopt governance structures that incentivize long-term value creation. For policymakers, our findings suggest leveraging market competition to achieve environmental goals. By fostering competition, they can indirectly incentivize firms to proactively invest in green strategies and secure first-mover advantages, promoting a more efficient transition.

The underlying mechanism of our findings—the strategic growth option theory—extends well beyond the Chinese context. As biodiversity loss becomes a global concern, these insights are highly relevant for firms and policymakers in other emerging markets and developed economies, guiding a transition toward more sustainable and resilient corporate strategies.

Statement: During the preparation of this work the authors used ChatGPT-4o (OpenAI) in order to enhance the manuscript's readability and language quality. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Table 1. Variable definitions.

Category	Variable	Symbol	Definition
Dependent variable	Corporate investment	Invest	Cash paid for the construction of fixed assets, intangible assets and other long-term assets scaled by total assets in the previous year
		Alt_Invest	Cash paid for the construction of fixed assets, intangible assets and other long-term assets minus net cash recovered from the disposal of fixed assets, intangible assets, and other long-term assets, scaled by total assets in the previous year
Independent variable	Biodiversity risk	Bio_Risk	The proportion of the word frequency of biodiversity in the annual report's total word frequency
		Alt_Bio_Risk1	The proportion of the number of Chinese characters related to biodiversity in the annual report to the total number of Chinese characters
		Alt_Bio_Risk2	Equal to 1 if biodiversity terms appear more than twice in the annual report, and 0 otherwise
Climate risk	Climate physical risk	Climate_Risk_P	Province-level climate physical risk index, sourced from Gou et al. (2024)
	Climate policy uncertainty	CPU_P	Province-level climate policy uncertainty index, sourced from Ma et al. (2024)
Firm-level controls	Tobin's Q	Tobin's Q	The ratio of the market value of assets to the book value of assets
	Sales growth	Sales_Growth	Growth rate of operating income
	Cash flow	Cashflow	The ratio of operating cashflow to total assets
	Irreversibility	Irreversibility	The ratio of fixed assets to total assets
	Firm size	Size	Natural logarithm of year-end total value
	Leverage	Leverage	Total liabilities/total assets
	Board male ratio	Male_Ratio	Number of male member divided by the total number of board
	Board age	Board_Age	Natural logarithm of average age of board members
Macro-level controls	Regional GDP	GDP_P	Natural logarithm of province-level GDP

Note: This table presents the definition of main variables.

Table 2. Descriptive statistics.

Variables	Obs	Mean	Std. Dev.	Min	Max
Invest	39,454	0.0606	0.0636	0.0003	0.3407
Alt_Invest	39,454	0.0581	0.0636	-0.0191	0.3368
Bio_Risk	39,454	0.0002	0.0004	0.0000	0.0029
Alt_Bio_Risk1	39,454	0.0001	0.0003	0	0.0020
Alt_Bio_Risk2	39,454	0.4937	0.5000	0.0000	1.0000
Climate_Risk_P	39,454	44.7794	7.4759	25.8066	84.3409
CPU_P	39,454	2.3945	0.7750	0.6512	5.4913
Tobin's Q	39,454	2.6044	1.8752	0.8717	12.1393
Sales_Growth	39,425	0.3325	0.9239	-0.7039	6.6210
Cashflow	39,454	-0.0888	0.2469	-1.0114	0.5535
Irreversibility	39,454	0.2149	0.1594	0.0021	0.6952
Size	39,454	22.2533	1.2975	19.8029	26.1990
Leverage	39,454	0.4288	0.2023	0.0508	0.8918
Male_Ratio	39,454	80.6209	11.4349	50.0000	100.0000
Board_Age	39,454	49.4017	3.1977	41.2900	56.7800
GDP_P	39,454	10.5771	0.8151	5.9785	11.8180

Note: This table presents the summary statistics of the main variables.

Table 3. Baseline regression.

Variables	Investment		
	(1)	(2)	(3)
Bio_Risk	4.9177** (2.32)	4.8854** (2.31)	7.0188*** (3.52)
Climate_Risk_P		-0.0002*** (-2.65)	-0.0001** (-2.55)
CPU_P		-0.0006 (-0.64)	-0.0009 (-0.94)
Tobin's Q			0.0043*** (12.48)
Sales_Growth			-0.0004 (-1.02)
Cashflow			-0.0242*** (-16.70)
Irrevisiblity			-0.0788*** (-11.14)
Size			-0.0111*** (-8.77)
Leverage			-0.0343*** (-8.43)
Male_Ratio			0.0001 (1.27)
Board_Age			0.0004* (1.78)
GDP_P			-0.0022 (-0.52)
Constant	0.0597*** (150.06)	0.0685*** (19.17)	0.3242*** (6.41)
Observations	39,454	39,454	39,454
R-squared	0.473	0.473	0.510
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Cluster by Firm	Yes	Yes	Yes

Note: This table presents the results regarding the impact of biodiversity risk on corporate investment. Firm and year fixed effects are included in all regressions. The *t*-values are included in parentheses, with standard errors clustered at the firm level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4. Mechanism analysis.

Variables	Investment					
	Industrial HHI		CR-4		CR-8	
	Low-competition (1)	High-competition (2)	Low-competition (3)	High-competition (4)	Low-competition (5)	High-competition (6)
Bio_Risk	5.8237** (2.16)	9.6440*** (3.15)	5.4716* (1.95)	9.9231*** (3.36)	5.2090* (1.85)	8.9215*** (3.05)
Climate_Risk_P	-0.0001 (-0.78)	-0.0001* (-1.85)	-0.0001 (-1.09)	-0.0001 (-1.45)	-0.0001* (-1.78)	-0.0000 (-0.64)
CPU_P	0.0001 (0.09)	-0.0002 (-0.11)	0.0001 (0.08)	-0.0002 (-0.14)	-0.0001 (-0.05)	-0.0001 (-0.10)
Tobin's Q	0.0039*** (7.37)	0.0045*** (10.10)	0.0040*** (7.09)	0.0044*** (9.90)	0.0040*** (7.04)	0.0043*** (9.87)
Sales_Growth	-0.0001 (-0.14)	-0.0005 (-0.96)	-0.0003 (-0.57)	-0.0004 (-0.72)	-0.0002 (-0.38)	-0.0002 (-0.43)
Cashflow	-0.0210*** (-10.45)	-0.0216*** (-10.79)	-0.0209*** (-9.82)	-0.0217*** (-11.36)	-0.0238*** (-11.10)	-0.0183*** (-9.39)
Irrevisiblity	-0.0717*** (-7.36)	-0.1157*** (-11.40)	-0.0697*** (-7.12)	-0.1065*** (-10.13)	-0.0695*** (-7.09)	-0.1219*** (-11.87)
Size	-0.0137*** (-6.89)	-0.0128*** (-6.84)	-0.0122*** (-6.21)	-0.0126*** (-6.84)	-0.0135*** (-6.86)	-0.0135*** (-7.06)
Leverage	-0.0363*** (-6.11)	-0.0294*** (-4.87)	-0.0402*** (-6.62)	-0.0239*** (-3.85)	-0.0382*** (-6.31)	-0.0239*** (-3.94)
Male_Ratio	0.0001 (0.91)	0.0001* (1.67)	0.0001 (0.73)	0.0001* (1.65)	0.0000 (0.27)	0.0002** (2.55)
Board_Age	0.0004 (1.09)	0.0004 (1.10)	0.0004 (1.04)	0.0003 (0.96)	0.0004 (0.93)	0.0004 (1.07)
GDP_P	-0.0024 (-0.41)	-0.0072 (-1.12)	-0.0001 (-0.02)	-0.0065 (-1.07)	-0.0003 (-0.04)	-0.0062 (-0.93)
Constant	0.3828*** (5.10)	0.4130*** (5.35)	0.3294*** (4.24)	0.4009*** (5.28)	0.3713*** (4.86)	0.4061*** (5.08)
Diff-P value	0.010		0.010		0.000	
Observations	19,349	19,423	17,499	21,331	18,730	20,277
R-squared	0.545	0.544	0.534	0.550	0.523	0.559
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster by Firm	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table presents the mechanisms of the effect of biodiversity risk on corporate investment. Industrial competition are proxied by three measures: the Herfindahl-Hirschman Index (HHI), the four-firm concentration ratio (CR4), and the eight-firm concentration ratio (CR8). High competition is coded as 1 (when the magnitude is below its mean) and low competition as 0. Control variables are defined in Table 1. All regressions include firm and year fixed effects. The *t*-values are reported in parentheses, with standard errors clustered at the firm level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 5. Further analysis for mechanism analysis.

Variables	Green Innovation Patent	Green Utility Model Patent
	(1)	(2)
Bio_Risk	189.2324*** (5.75)	151.2053*** (5.02)
Climate_Risk_P	-0.0012* (-1.65)	0.0005 (0.71)
CPU_P	0.0122 (0.95)	0.0231* (1.95)
Tobin's Q	0.0156*** (4.78)	0.0119*** (4.01)
Sales_Growth	-0.0043 (-0.92)	-0.0035 (-0.82)
Cashflow	-0.0173 (-1.22)	-0.0205 (-1.49)
Irrevisiblity	0.0576 (0.77)	0.2592*** (3.77)
Size	0.2572*** (14.19)	0.2022*** (12.53)
Leverage	-0.1092** (-2.17)	-0.0536 (-1.12)
Male_Ratio	0.0022** (2.51)	0.0017** (2.20)
Board_Age	0.0021 (0.65)	0.0053* (1.70)
GDP_P	0.0227 (0.39)	-0.0932* (-1.65)
Constant	-5.6424*** (-7.65)	-3.5008*** (-5.15)
Observations	39,427	39,427
R-squared	0.717	0.698
Firm FE	Yes	Yes
Year FE	Yes	Yes
Cluster by Firm	Yes	Yes

Note: This table extends the mechanism test by examining the relationship between biodiversity risks and the number of green patents and utility model held by firms. Control variables are defined in Table 1. All regressions include firm and year fixed effects. The *t*-values are reported in parentheses, with standard errors clustered at the firm level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6. Robustness check.

Variables	Alternative Biodiversity_Risk		Alternative Investment	Entropy Balance	2SLS		DID
	Invest	Invest	Alt_Invest	Invest	First Stage	Second Stage	Invest
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Alt_Bio_Risk1	10.2349*** (3.62)						
Alt_Bio_Risk2		0.0036*** (4.11)		0.0026*** (3.13)			
Bio_Risk			6.5967*** (3.27)			31.0634** (1.98)	
Industry_Bio_IV1					0.3032*** (6.10)		
Forest_Rate_IV2					0.0002** (2.35)		
DID							0.0132*** (2.72)
Climate_Risk_P	-0.0001** (-2.55)	-0.0002*** (-2.63)	-0.0001** (-2.45)	-0.0001 (-1.59)	-0.0000 (-1.00)	-0.0001** (-2.34)	-0.0002** (-2.56)
CPU_P	-0.0009 (-0.94)	-0.0010 (-0.99)	-0.0009 (-0.96)	-0.0008 (-0.81)	0.0000 (0.89)	-0.0015 (-1.48)	-0.0013 (-1.25)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	39,454	39,454	39,454	39,454	33,503	33,503	33,654
R-squared	0.511	0.510	0.507	0.556	-	0.0647	0.553
Kleibergen-Paap					23.13		
Wald F-statistic	-	-	-	-	(10% max IV size = 19.93)		-
Hansen J statistic	-	-	-	-	Chi-sq P-value = 0.2154		-

Note: This table presents the results of various robustness tests, including the substitution of explanatory variables, the dependent variable, and the control for endogeneity. Columns (1) and (2) use two alternative independent variables. *Alt_Bio_Risk1* is the total count of biodiversity-related keywords in a firm's annual report. *Alt_Bio_Risk2* is a binary indicator equal to 1 if more than two biodiversity-related terms appear, and 0 otherwise. Column (3) introduces an alternative dependent variable, *Alt_Invest*, calculated as the cash paid for the construction of fixed, intangible, and other long-term assets minus the cash received from the disposal of such assets, scaled by lagged total assets. Column (4) perform entropy balance method. Column (5)-(6) reports results from the 2SLS regression using two instrumental variables: *Industry_Bio_IV1*, defined as the average biodiversity risk of same-industry firms located in other provinces; and *Forest_Rate_IV2*, the province-level forest coverage rate. Column (7) perform DID analysis. All regressions include firm and year fixed effects. The *t*-values are reported in parentheses, with standard errors clustered at the firm level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 7. Heterogeneity analysis.

Variables	Investment			
	Climate policy uncertainty		Managerial myopia	
	Low (1)	High (2)	Low (3)	High (4)
Bio_Risk	5.4228* (1.75)	9.6412*** (3.42)	7.3877*** (3.01)	4.5856 (1.51)
Climate_Risk_P	-0.0001 (-1.51)	-0.0002*** (-2.74)	-0.0001* (-1.66)	-0.0001 (-1.62)
CPU_P	-0.0006 (-0.44)	0.0015 (0.98)	-0.0003 (-0.26)	-0.0004 (-0.26)
Tobin's Q	0.0044*** (8.22)	0.0041*** (9.20)	0.0044*** (10.98)	0.0047*** (7.06)
Sales_Growth	-0.0009 (-1.58)	-0.0001 (-0.10)	-0.0003 (-0.55)	-0.0001 (-0.24)
Cashflow	-0.0247*** (-10.94)	-0.0238*** (-11.91)	-0.0221*** (-12.31)	-0.0267*** (-11.73)
Irrevisiblity	-0.0685*** (-6.49)	-0.0953*** (-10.47)	-0.0846*** (-9.96)	-0.0722*** (-6.76)
Size	-0.0132*** (-7.02)	-0.0111*** (-6.37)	-0.0141*** (-8.75)	-0.0097*** (-5.07)
Leverage	-0.0369*** (-6.20)	-0.0344*** (-5.88)	-0.0298*** (-5.73)	-0.0430*** (-6.59)
Male_Ratio	0.0001 (0.53)	0.0001 (1.49)	0.0002** (2.06)	-0.0001 (-1.19)
Board_Age	0.0011*** (3.11)	-0.0002 (-0.56)	0.0003 (1.12)	0.0004 (1.18)
GDP_P	0.0018 (0.28)	-0.0042 (-0.63)	0.0047 (0.91)	-0.0048 (-0.85)
Constant	0.2990*** (4.08)	0.3766*** (4.64)	0.3117*** (4.82)	0.3410*** (4.84)
Diff-P value	0.010		0.050	
Observations	17,777	20,266	23,518	14,881
R-squared	0.536	0.558	0.552	0.547
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Cluster by Firm	Yes	Yes	Yes	Yes

Note: This table presents the heterogeneity analysis of the effect of biodiversity risk on corporate investment. The observations are separately categorized into high and low groups based on the mean values of the province-level climate policy uncertainty index and the managerial myopia index. Control variables are defined in Table 1. All regressions include firm and year fixed effects. The *t*-values are reported in parentheses, with standard errors clustered at the firm level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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